Muhammad Moiz Bukhari

Software and Systems Engineer · Python, C/C++, C#, SQL · Test Automation and FDA-Regulated V&V
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Experience

Systems Engineering Co-op Jan 2025 – Aug 2025
Abbott Laboratories *Pleasanton, CA*

Test Software, Automation, and Debugging

- Eliminated manual overnight test monitoring by designing a **5-fixture** power automation system pairing embedded C/C++ firmware with **Python test sequencing** to run **72-hour** unattended endurance cycles.
- Became the go-to engineer for cross-team debug and root cause by building **Python test infrastructure** with real-time logging and anomaly detection across **4+ parallel** fixture configurations, alongside hardware and quality teams.
- Reduced rework across **5 fixture revisions** by structuring the codebase into separate sensor-driver, relay-control, and communication layers, so each new build changed one module instead of the whole program.
- Improved reliability of automated test cycles by tracing and fixing state-machine bugs through serial-monitor logs and live LCD diagnostics, iterating on hardware-in-the-loop behavior until cycles ran clean.
- Delivered builds to Long-Term Testing without rework cycles by owning the fixture end to end, from firmware and relay/sensor integration to wiring, soldering, and continuity/voltage checks against schematics.

Design Controls, Verification, and Quality System

- Verified secure BLE communication on a **Class III cardiac device program** by executing formal V&V protocols across multiple pairing scenarios, documenting results with traceability to design inputs and system requirements.
- Released **4 test fixtures** for V&V and Long-Term Testing by authoring IQ/OQ validation packages, PMs, and drawings as change orders in **Windchill PLM** and working them through CIB approval under **FDA 21 CFR 820** and **ISO 13485**.
- Kept change records audit-ready by applying Good Documentation Practice to every CO/CA, including pulling back and reworking orders until they met reviewer and QMS requirements.
- Supported **3 summative human factors studies** across 3 sites for next-generation cardiac device systems, working with clinical and human factors engineers and troubleshooting equipment live during patient and clinician sessions.

Software Engineer May 2026 – Oct 2026
TWM Wholesale *Stockton, CA*

- Designed and built a full e-commerce platform from scratch, including user authentication, server-side data encryption, and a relational product database supporting a **40,000+ SKU** catalog.
- Saved **3–4 hours/day** of manual data entry across inventory and procurement with **10+ Python** automation scripts.
- Raised automation uptime by tracing recurring failures through log analysis and rebuilding scripts as fault-tolerant pipelines.

Projects

WaterCon – Cloud-Connected IoT Platform Python, REST, SQL

- Cut stakeholder reporting lag from hours to seconds by designing **RESTful APIs** and a normalized **SQL** schema for real-time telemetry from 3+ sensor device types.
- Sustained **99.5%** platform uptime by building fault-tolerant Python ingestion pipelines with automated anomaly alerts.

End-to-End Encrypted Messenger (Team Project) TypeScript, React Native, Go

- Built a two-person mobile messenger on the Signal Protocol (X3DH, Double Ratchet) with a React Native/Expo client, a Go backend, WebSocket delivery with auto-reconnect, and safety-number identity verification.

Skills

Languages: Python, C#, C/C++, SQL, Java, JavaScript, TypeScript, Go, Bash, PowerShell, MATLAB

Software & Data: Object-oriented design, RESTful API design, relational database design, query optimization, ETL pipelines, test automation, debugging and root-cause analysis

Regulated Development: Design controls, FDA 21 CFR 820, ISO 13485, IEC 62304, V&V protocols, IQ/OQ/PQ, requirements traceability, Good Documentation Practice, Windchill PLM

Tools & Process: Git/BitBucket, Linux, VMware, LabVIEW, Arduino, BLE, Agile/Scrum (Certified Scrum Master)

Education

B.S. Computer Science, Honors – University of the Pacific, Stockton, CA Aug 2021 – Dec 2025
GPA 3.83 · Dean's List, all semesters